

05_Simple_Switch_Configuration – Configure VLANs, Assign Ports, Test Communication

Packet Tracer – Simple Switch VLAN Configuration

Objectives

Part 1: Build a Network with a Switch

Part 2: Create VLANs on the Switch

Part 3: Assign Switch Ports to VLANs

Part 4: Configure Static IP Addressing on PCs

Part 5: Test Communication Within and Across VLANs

Background / Scenario

You are learning how a switch can logically divide a network using **Virtual Local Area Networks (VLANs)**. VLANs allow devices connected to the same physical switch to be separated into different broadcast domains.

In this lab, you will:

- Connect four PCs to a switch
- Create two VLANs on the switch
- Assign specific switch ports to each VLAN
- Configure static IPv4 addresses on PCs
- Test communication between devices in the same VLAN and different VLANs

This lab reinforces understanding of:

- Basic switch configuration
- VLAN creation and port assignment
- Network segmentation
- Same-VLAN communication
- Why devices in different VLANs cannot communicate without a router or Layer 3 device

Instructions

Part 1: Connect the Devices

Step 1: Add Devices

- a. Open Cisco Packet Tracer.
- b. From **End Devices**, drag the following onto the workspace:

- PC0
- PC1
- PC2
- PC3

c. From **Network Devices** → **Switches**, drag **Switch0** onto the workspace.

d. Rename the PCs as follows:

- PC0 → **PC-A**
 - PC1 → **PC-B**
 - PC2 → **PC-C**
 - PC3 → **PC-D**
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Step 2: Connect the Devices

a. Click **Connections** (lightning bolt icon).

b. Use **Copper Straight-Through** cables for all connections.

c. Connect the PCs to **Switch0** using these exact ports:

- **PC-A FastEthernet0 → Switch0 FastEthernet0/1**
- **PC-B FastEthernet0 → Switch0 FastEthernet0/2**
- **PC-C FastEthernet0 → Switch0 FastEthernet0/3**
- **PC-D FastEthernet0 → Switch0 FastEthernet0/4**

d. Wait for the link lights to turn green.

Your physical topology is now complete.

Part 2: Create VLANs on the Switch

Step 1: Access the Switch CLI

a. Click **Switch0**.

b. Go to the **CLI** tab.

c. Press **Enter** if prompted.

Step 2: Create VLAN 10 and VLAN 20

a. Enter the following commands:

```
enable
configure terminal
vlan 10
name SALES
exit
vlan 20
name IT
exit
```

Explanation

- **VLAN 10** will be used for **PC-A** and **PC-B**
 - **VLAN 20** will be used for **PC-C** and **PC-D**
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Part 3: Assign Switch Ports to VLANs

Step 1: Assign Ports for VLAN 10

a. In the same CLI, enter:

```
interface range fastEthernet 0/1 - 2
switchport mode access
switchport access vlan 10
exit
```

Explanation

This assigns:

- **FastEthernet0/1** → VLAN 10
- **FastEthernet0/2** → VLAN 10

So **PC-A** and **PC-B** will belong to VLAN 10.

Step 2: Assign Ports for VLAN 20

a. Enter the following commands:

```
interface range fastEthernet 0/3 - 4
switchport mode access
switchport access vlan 20
exit
```

end
write memory

Explanation

This assigns:

- **FastEthernet0/3** → VLAN 20
- **FastEthernet0/4** → VLAN 20

So **PC-C** and **PC-D** will belong to VLAN 20.

Part 4: Configure Static IP Addressing on PCs

Step 1: Configure PC-A

a. Click **PC-A** → **Desktop** → **IP Configuration**

b. Enter:

- **IP Address:** 192.168.10.1
 - **Subnet Mask:** 255.255.255.0
 - **Default Gateway:** leave blank
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Step 2: Configure PC-B

a. Click **PC-B** → **Desktop** → **IP Configuration**

b. Enter:

- **IP Address:** 192.168.10.2
 - **Subnet Mask:** 255.255.255.0
 - **Default Gateway:** leave blank
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Step 3: Configure PC-C

a. Click **PC-C** → **Desktop** → **IP Configuration**

b. Enter:

- **IP Address:** 192.168.20.1
 - **Subnet Mask:** 255.255.255.0
 - **Default Gateway:** leave blank
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Step 4: Configure PC-D

a. Click **PC-D** → **Desktop** → **IP Configuration**

b. Enter:

- **IP Address:** 192.168.20.2
- **Subnet Mask:** 255.255.255.0
- **Default Gateway:** leave blank

Explanation

- PCs in **VLAN 10** use the **192.168.10.0/24** network
 - PCs in **VLAN 20** use the **192.168.20.0/24** network
 - Since there is no router in this lab, **inter-VLAN communication will not work**
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Part 5: Test Communication

Step 1: Ping PC-B from PC-A

a. Click **PC-A** → **Desktop** → **Command Prompt**

b. Type:

```
ping 192.168.10.2
```

Expected Result

You should receive **4 successful replies** because **PC-A** and **PC-B** are in the same VLAN.

Step 2: Ping PC-D from PC-C

a. Click **PC-C** → **Desktop** → **Command Prompt**

b. Type:

```
ping 192.168.20.2
```

Expected Result

You should receive **4 successful replies** because **PC-C** and **PC-D** are in the same VLAN.

Step 3: Ping PC-C from PC-A

a. On **PC-A**, type:

```
ping 192.168.20.1
```

Expected Result

The ping should **fail** because **PC-A** and **PC-C** are in different VLANs, and there is no router or Layer 3 switch to route traffic between them.

Step 4: Ping PC-A from PC-D

a. On **PC-D**, type:

```
ping 192.168.10.1
```

Expected Result

The ping should **fail** for the same reason: different VLANs cannot communicate directly through a Layer 2 switch.

Verification Commands on the Switch

To verify VLAN creation and port assignments, go to **Switch0 CLI** and enter:

```
show vlan brief
```

Expected Result

You should see something similar to:

VLAN Name	Status	Ports
1 default	active	
10 SALES	active	Fa0/1, Fa0/2
20 IT	active	Fa0/3, Fa0/4

This confirms that the ports were assigned correctly.

Troubleshooting Tips

✓ Make sure all PC-to-switch connections use **Copper Straight-Through** cables.

✓ Verify each PC is connected to the correct switch port:

- PC-A → Fa0/1
- PC-B → Fa0/2
- PC-C → Fa0/3
- PC-D → Fa0/4

- ✓ Confirm the switch ports were assigned to the correct VLANs.
 - ✓ Check IP addresses and subnet masks carefully.
 - ✓ Use show vlan brief to confirm VLAN membership.
 - ✓ Remember: devices in different VLANs cannot communicate without a router or Layer 3 switch.
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What You Learned

- How to create VLANs on a switch
 - How to assign switch ports to specific VLANs
 - How VLANs separate devices into different logical networks
 - How devices in the same VLAN can communicate normally
 - Why inter-VLAN communication requires routing
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Challenge Extensions (Optional)

1. Add a router and configure **router-on-a-stick** for inter-VLAN communication.
2. Add another switch and configure a **trunk link** between switches.
3. Create a third VLAN and assign another port to it.
4. Use show running-config to inspect the switch configuration.